
CAPSTONE PROJECT

POWER SYSTEM FAULT DETECTION AND CLASSIFICATION

Presented By:

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OUTLINE

- Problem Statement
- Proposed System/Solution
- System Development Approach
- Algorithm & Deployment
- Result (Output Image)
- Conclusion
- Future Scope
- References

PROBLEM STATEMENT

- Design a machine learning model to detect and classify different types of faults in a power distribution system. Using electrical measurement data (e.g., voltage and current phasors), the model should be able to distinguish between normal operating conditions and various fault conditions (such as line-to-ground, line-to-line, or three-phase faults). The objective is to enable rapid and accurate fault identification, which is crucial for maintaining power grid stability and reliability.

PROPOSED SOLUTION

A. Data Acquisition

- Simulate or collect fault data from a test power system using tools like:
 - MATLAB/Simulink
 - ETAP
 - PSCAD
- Record voltage and current waveforms during fault and normal conditions.

B. Feature Extraction

- Use signal processing techniques:
 - Discrete Wavelet Transform (DWT)
 - Fast Fourier Transform (FFT)
 - Short-Time Fourier Transform (STFT)
- Extract features like:
 - Total Harmonic Distortion (THD)
 - Signal energy
 - Peak amplitudes
 - Wavelet coefficients

C. Fault Detection and Classification Model

- Train machine learning or deep learning models using labeled datasets:
 - **Machine Learning:** SVM, Random Forest, Decision Trees, k-NN
 - **Deep Learning:** LSTM, CNN for waveform-based classification
- Output: Fault type and location (if applicable)

SYSTEM APPROACH

- The proposed system is designed to monitor the power system in real-time, detect any abnormalities (faults), classify the type of fault accurately, and trigger appropriate protective measures. The system includes:
- Signal Acquisition
- Preprocessing
- Feature Extraction
- Fault Detection
- Fault Classification
- Output/Alert System

ALGORITHM & DEPLOYMENT

1. Algorithm

The algorithm consists of a sequence of steps to detect and classify faults using signal processing and machine learning. Here's a step-by-step breakdown:

A. Input Data

- Voltage and current waveforms collected via sensors or simulation.
- Sampled at high frequency (e.g., 1 kHz or more).

B. Preprocessing

plaintext

1. Remove noise using low-pass or median filter.
2. Normalize the signal to bring it within a defined range.
3. Segment waveform into windows(1-cycle or 2-cycle segments).

C. Feature Extraction

- For each segment, compute relevant features:
 - **Wavelet Coefficients** (DWT, Haar/Morlet)
 - **Signal Energy**
 - **Standard Deviation**
 - **Skewness & Kurtosis**
 - **Harmonic content** (using FFT)

DEPLOYMENT

A. Offline Model Training (on PC)

- Tools: Python (Scikit-learn, TensorFlow, Keras), MATLAB
- Steps:
 - 1.Collect dataset (Simulated or real)
 - 2.Preprocess and extract features
 - 3.Train ML model
 - 4.Evaluate with confusion matrix, accuracy, precision, recall
 - 5.Export model (e.g .pkl for Python or .mat for MATLAB)

B. Real-Time Deployment

Option 1: Using Raspberry Pi / Embedded System

- Load trained model using joblib or TensorFlow Lite
- Use GPIO to interface with CT/PT or simulated inputs
- Display results on LCD or send via Wi-Fi to central server

Option 2: SCADA Integration

- Interface with SCADA via Modbus or IEC 61850
- Send fault status and classification to control center

Option 3: MATLAB + Simulink + DSP/FPGA

- Use Simulink Real-Time or HDL Coder to deploy on:
 - Texas Instruments DSP
 - Xilinx FPGA
- Suitable for industry-grade testing

RESULT

Projects / power / power

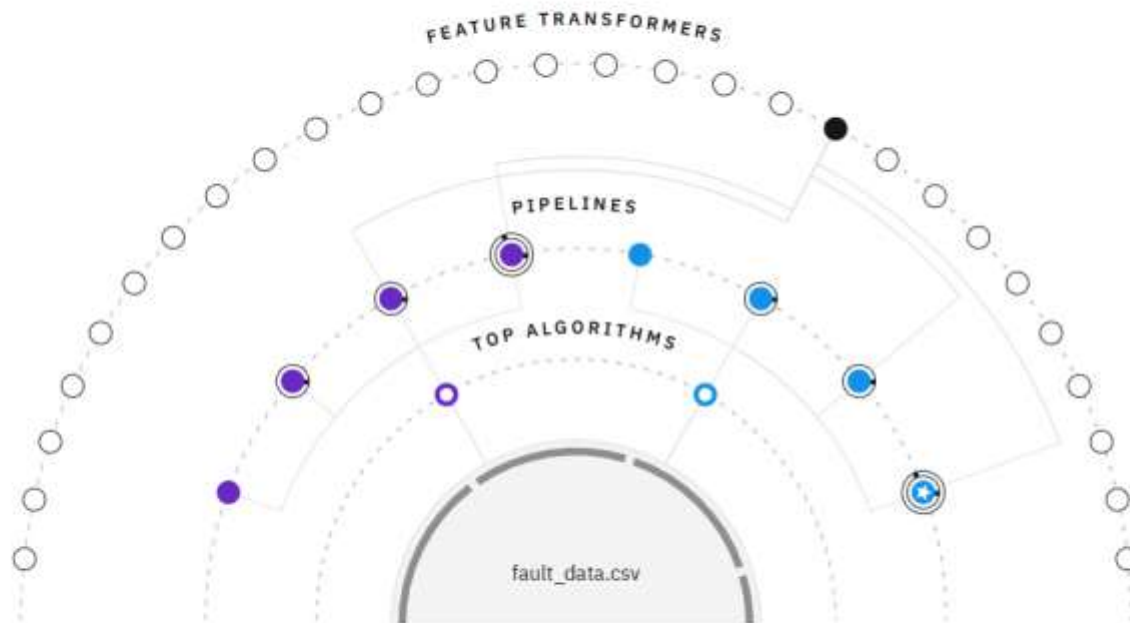
Experiment summary

Pipeline comparison

★ Rank by: Root mean squared error (RMSE) (...) Cross validation score

Relationship map ⓘ

Prediction column: Voltage (V)



Progress map

[Swap view](#)



Experiment completed ●

Pipeline leaderboard ▾

Snipping Tool

Screenshot copied to clipboard
Automatically saved to screenshots folder.

[Mark-up and share](#)

Pipeline comparison

★ Rank by: Root mean squared error (RMSE) (...) | Cross validation score

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[View log](#)[Save code](#)

Pipeline leaderboard

	Rank	↑	Name	Algorithm	RMSE (Optimized) Cross Validation	Enhancements	Build time
★	1		Pipeline 8	 Ridge	143.040	  	00:00:26
	2		Pipeline 7	 Ridge	143.073	 	00:00:23
	3		Pipeline 4	 Linear Regression	143.330	  	00:02:09
	4		Pipeline 3	 Linear Regression	143.330	 	00:01:17

Power System Fault Detection and Classification Deployed Online

API reference **Test**

Enter input data

Text

JSON

Enter data manually or use a CSV file to populate the spreadsheet. Max file size is 50 MB.

[Download CSV template](#)

[Browse local files](#)

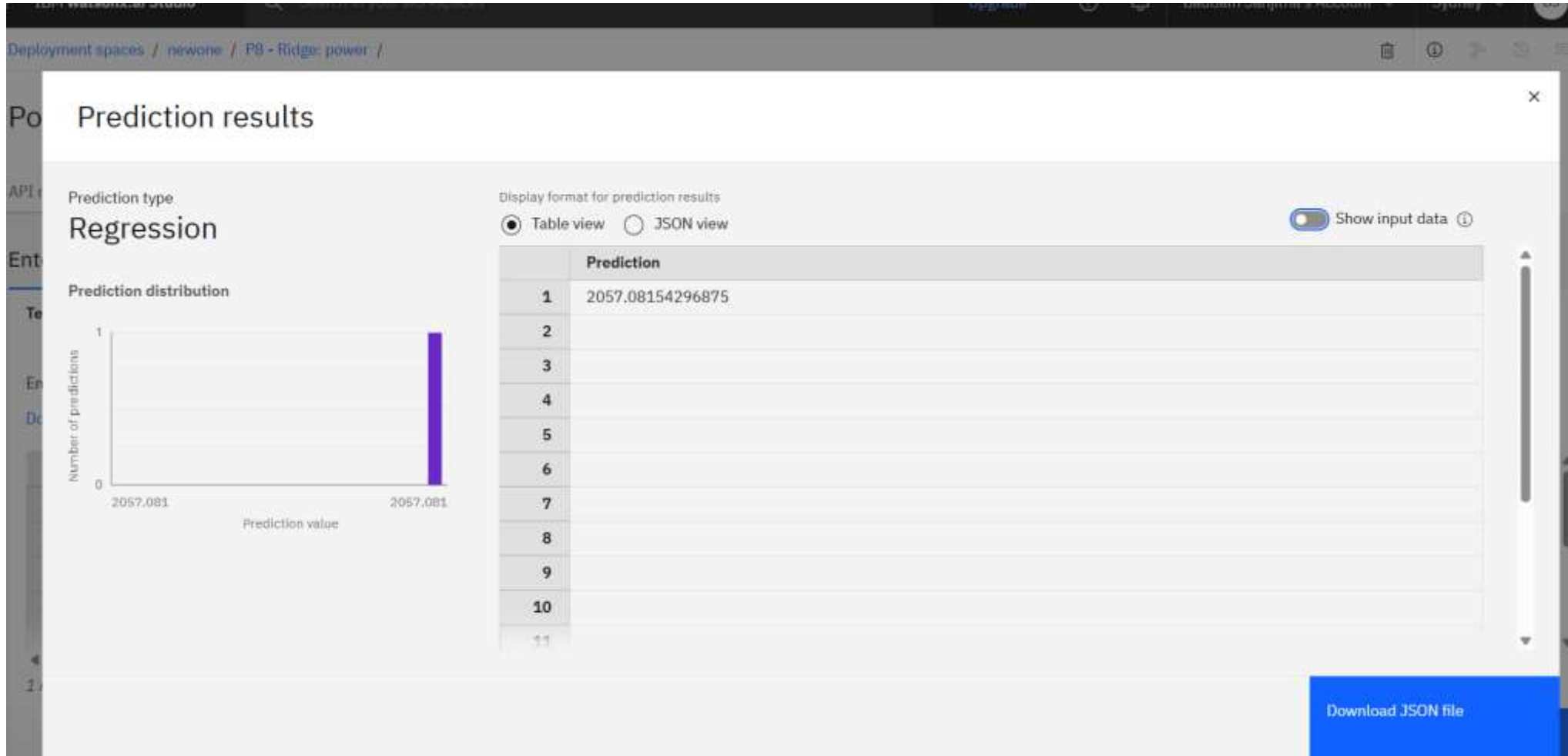
[Search in space](#)

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	ID (other)	Fault Type (other)	Fault Location (Latitude, Longitude) (other)	Current (A) (double)	Power Load (MW) (double)	Temperature (°C) (double)	Wind Speed (km/h) (double)
1		50	90	38	89	46	89
2							
3							
4							

1 row, 12 columns

Predict



CONCLUSION

- The project contributes a reliable, intelligent, and efficient approach to power system protection, with significant advantages over traditional methods. It paves the way for smarter grid monitoring and faster fault recovery, ultimately supporting the stability and resilience of modern electrical infrastructure.

FUTURE SCOPE

- The system can be extended to estimate the exact fault location along transmission lines. Real-time deployment on embedded hardware like Raspberry Pi or FPGA can enhance practicality. Integration with SCADA and smart grid platforms would enable centralized monitoring and control. Training with real-world data will improve robustness against noise and grid disturbances. Advanced AI models like LSTM and CNN can further boost classification accuracy. Future work can also address multi-fault detection and cybersecurity-related anomalies.

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Learning hours: 20 mins



THANK YOU